ECON 770 Recitation 10

Oct 16, 2020

University of North Carolina at Chapel Hill

Reminder

- Due of Assignment #4.2: Oct 20, 2020 12:00 PM

Contents

- Sampling, Statistics and Estimator
- Unbiasedness
- Efficient Estimator

i.i.d. Sampling

Theorem

If sample $\{X_i\}_{i=1}^n$ are i.i.d. draws, then for any function $g(\cdot)$, $\{g(X_i)\}_{i=1}^n$ are i.i.d.

Example: Let $\{X_i\}_{i=1}^n \sim$ i.i.d. $N(0,1)$, then $\{X_i^2\}_{i=1}^n \sim$ i.i.d. $\chi^2(1)$

i.i.d. Sampling

Defn.

Given data $\{X_i\}_{i=1}^n$, a statistic $\hat{\theta} \in \mathbb{R}^q$ (where $q < n$) is a function of data, i.e.

$$\hat{\theta} = g(X_1, \ldots, X_n) : \ \mathbb{R}^n \to \mathbb{R}^q$$

- $\hat{\theta}$ is a point estimator
- $\hat{\theta}$ is a random variable with distribution

Example: $\{X_i\}_{i=1}^n = [1, 4, -2]'$, $\theta \equiv \mathbb{E}[X_i]$, $\hat{\theta} = \frac{1}{n}\sum_{i=1}^n X_i = 1$

Statistics and Estimator

Defn. (Statistics)

Given data $\{X_i\}_{i=1}^n$,

- Sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$
- Sample mean of function $\overline{g(X)} = \frac{1}{n}\sum_{i=1}^n g(X_i)$
- Biased sample variance $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$
- Unbiased sample variance $s_x^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$

Defn. (Estimators)

- Plug-in estimator $\beta = h(\theta)$ where $\theta = \mathbb{E}[g(X)]$

Example: Let $g(x) = [x, x^2]$, $h([x_1, x_2]) = x_2 - x_1^2$, then

$$\sigma^2 = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 = h(\mathbb{E}[X], \mathbb{E}\left[X^2\right]) = h(\mathbb{E}[g(x)]),\ \hat{\sigma}^2 = h(\bar{X}, \bar{X^2})$$

Bias and Unbiasedness

Defn.

Given statistics $\hat{\theta}$,

- $Bias(\hat{\theta}) = \mathbb{E}\left[\hat{\theta}\right] - \theta$
- $\hat{\theta}$ is unbiased if $Bias(\hat{\theta}) = 0$, i.e. $\mathbb{E}\left[\hat{\theta}\right] = \theta$.

Example

- $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ where $X_i \sim i.i.d.\ F(\mu)$, $\mathbb{E}\left[\bar{X}\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i\right] = \mu$
- $\hat{\mu} = \sum_{i=1}^{n} w_i X_i$ where $w_i \geq 0$ and $\sum_{i=1}^{n} w_i = 1$, $\mathbb{E}\left[\hat{\mu}\right] = \sum_{i=1}^{n} w_i \mu = \mu$

Bias and Nonlinear Transformation

It's not necessary to have $h\left[\mathbb{E}(\hat{\theta})\right] = \mathbb{E}\left[h(\hat{\theta})\right]$ if $\mathbb{E}\left[\hat{\theta}\right] = \theta$.

Theorem

- If $h(\theta) = a + b\theta$ and $\mathbb{E}\left[\hat{\theta}\right]$, then $h\left[\mathbb{E}(\hat{\theta})\right] = \mathbb{E}\left[h(\hat{\theta})\right] = a + b\theta$
- If $h(\theta)$ is strictly concave and $\mathbb{E}\left[\hat{\theta}\right]$, then $h\left[\mathbb{E}(\hat{\theta})\right] > \mathbb{E}\left[h(\hat{\theta})\right]$

Proof by linearity and Jensen's Inequality.

Exercise 1

1. Let $X_i \sim i.i.d.\ (\mu, \sigma^2)$, show $s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is an unbiased estimator of σ^2
2. Let $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$, prove that $Bias(\hat{\sigma}^2) \longrightarrow 0$ when $n \longrightarrow \infty$.

Mean Squared Error (MSE)

Measure of error for possible biased estimators

Defn.

Let $\hat{\theta}$ be the estimator of θ, then

$$MSE \equiv \mathbb{E}\left[(\hat{\theta} - \theta)^2\right]$$

which equals to $V(\hat{\theta})$ if $\mathbb{E}[\hat{\theta}] = \theta$

Inequalities

Let $MSE(\hat{\theta}) < \infty$,

- Chebyshev's Inequality: for any $\varepsilon > 0$,

$$P(|\hat{\theta} - \theta| > \varepsilon) \leq \frac{1}{\varepsilon^2}\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \frac{1}{\varepsilon^2}MSE(\hat{\theta})$$

- Markov's Inequality: for any $\varepsilon, p > 0$, if $\mathbb{E}\left[|\hat{\theta} - \theta|^p\right] < \infty$, then

$$P(|\hat{\theta} - \theta| > \varepsilon) \leq \frac{1}{\varepsilon^p}\mathbb{E}\left[|\hat{\theta} - \theta|^p\right]$$

(How to prove Chebyshev's Inequality?)

Best Linear Unbiased Estimator

Here "Best" means the estimator is **efficient**, i.e. the estimator has the lowest variance.

Defn.

If a linear estimator $\hat{\theta}$ is unbiased with the least variance, then it is the Best Linear Unbiased Estimator (BLUE).

Example: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ has the smallest variance among $\hat{\mu} = \sum_{i=1}^{n} w_i X_i$ (where $\sum_{i=1}^{n} w_i = 1$ s.t. the estimator is unbiased). Proof by solving

$$\min_{w_i} Var[\hat{\mu}] \text{ s.t. } \sum_{i=1}^{n} w_i = 1$$

Example: Gauss-Markov Theorem says OLS estimator is BLUE given proper assumptions

Min. Variance Unbiased Estimator

What if we want to find an unbiased estimator with least variance? This is hard because we may have thousands of unbiased estimators.

Theorem (Cramér - Rao Lower Bound)

Let $\theta \in \mathbb{R}^k$ be the unknown deterministic parameter. Denote $\{x_1, \ldots, x_n\}$ as n independent observations of x, where $x \sim f(x; \theta)$, and hence let joint density $(x_1, \ldots, x_n) \sim f_n(x; \theta)$. Then, every unbiased estimator $\hat{\theta}_n = g(x_1, \ldots, x_n)$ satisfies

$$Var[\hat{\theta}_n] = \left(\mathbb{E}\left[\frac{1}{f_n(x;\theta)^2} \frac{\partial f_n(x;\theta)}{\partial \theta} \frac{\partial f_n(x;\theta)}{\partial \theta'} \right] \right)^{-1} + D \equiv LB + D$$

where D is positive semi-definite.

When $\theta \in \mathbb{R}$, then $Var[\hat{\theta}_n] \geq CRLB \equiv \left(\mathbb{E}\left[\frac{1}{f_n(x;\theta)^2} \left(\frac{\partial f_n(x;\theta)}{\partial \theta} \right)^2 \right] \right)^{-1}$

Min. Variance Unbiased Estimator (Contd)*

- Here's another representation after we learn the likelihood function, i.e.

$$Var[\hat{\theta}_n] \geq \frac{1}{I(\theta)}$$

where $I(\theta) = n\mathbb{E}\left[\left(\frac{\partial l_n(x;\theta))}{\partial\theta}\right)^2\right]$ is called Fisher information[1] and $l_n(x;\theta) = \ln(f_n(x;\theta))$ is called the natural log-likelihood function.

- When the estimator $T(X)$ is biased, let $\phi(\theta) = \mathbb{E}[T(X)]$ and we know $\phi(\theta) - \theta \neq 0$. In this case, *CRLB* is given by

$$Var(T(X)) \geq CRLB = \frac{[\phi'(\theta)]^2}{I(\theta)}$$

[1] Here it is different from our lecture slides, because in that one we don't have n in $I(\theta)$, but instead we have n in CRLB

Min. Variance Unbiased Estimator (Example 1)

Suppose $X_i \sim i.i.d\ N(\mu, \sigma^2)$ where σ^2 is known, then $\theta = (\mu, \sigma^2) = \mu$ and

$$f_n(x;\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^2}\right)$$

$$\frac{\partial}{\partial\mu} f_n(x;\theta) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{n}\left(x_i-\mu\right)^2}{\sigma^2}\right) \times \frac{\sum_{i=1}^{n}\left(x_i-\mu\right)}{\sigma^2}$$

$$= f_n(x;\theta)\frac{\sum_{i=1}^{n}\left(x_i-\mu\right)}{\sigma^2}$$

$$\Rightarrow \quad CRLB^{-1} = \mathbb{E}\left[\left(\frac{\sum_{i=1}^{n}(x_i-\mu)}{\sigma^2}\right)^2\right] = \frac{n}{\sigma^2}$$

Therefore, $CRLB = \dfrac{\sigma^2}{n}$, which is exactly identical to $Var(\bar{X})$.

Min. Variance Unbiased Estimator (Example 2)

Suppose $X_i \sim i.i.d\ N(\mu, \sigma^2)$ where μ is known, then $\theta = (\mu, \sigma^2) = \sigma^2$. It is obvious that

$$T(X) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$$

is an unbiased estimator of σ^2. It can be shown that $Var(T(X)) = \frac{2\sigma^4}{n}$. We can also show that (proof in next page, but try to do it at first) the CRLB of unbiased estimator of σ^2 is $\frac{2\sigma^4}{n}$.

Min. Variance Unbiased Estimator (Proof)

Recall that

$$f_n(x;\theta) = \prod_{i=1}^{n} f(x_i;\theta) = \frac{1}{(2\pi)^{n/2}}(\sigma^2)^{-n/2}\exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^2}\right)$$

$$\begin{aligned}\frac{\partial}{\partial\sigma^2}f_n(x;\theta) &= \frac{1}{(2\pi\sigma^2)^{n/2}}\exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{n}\left(x_i-\mu\right)^2}{\sigma^2}\right)\times\left(-\frac{n}{2\sigma^2}\right)\\ &+ \frac{1}{(2\pi\sigma^2)^{n/2}}\exp\left(-\frac{1}{2}\frac{\sum_{i=1}^{n}\left(x_i-\mu\right)^2}{\sigma^2}\right)\times\left(\frac{1}{2}\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^4}\right)\\ &= f_n(x;\theta)\times\left(\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^4}-\frac{n}{2\sigma^2}\right)\end{aligned}$$

As a result $CRLB = \mathbb{E}\left[\frac{1}{4\sigma^4}\left(\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2-n\right)^2\right]^{-1}$

Min. Variance Unbiased Estimator (Proof, Contd)

What is $\mathbb{E}\left[\left(\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2-n\right)^2\right]$? Since $X_i \sim$ *i.i.d* $N(\mu,\sigma^2)$, let $Y=\sum_{i=1}^{n}(\frac{x_i-\mu}{\sigma})^2$ and then $Y\sim\chi^2(n)$.

Recall that $\mathbb{E}\left[Y^2\right]=n(n+2)=n^2+2n$ and $\mathbb{E}\left[Y\right]=n$,

$$\mathbb{E}\left[\left(\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2-n\right)^2\right]=\mathbb{E}\left[Y^2-2nY+n^2\right]$$

$$=n^2+2n-2n\times n+n^2=2n$$

Therefore,

$$CRLB=\frac{4\sigma^4}{2n}=\frac{2\sigma^4}{n}$$

Bias or Efficiency?

- Suppose that we are estimating σ^2 by using $X_i \sim \ i.i.d\ N(\mu, \sigma^2)$ given known μ
- Smaller variance but biased

$$T(X) = \frac{\sum_{i=1}^{n}(X_i - \mu)^2}{n+2} \quad \text{with} \quad Var(T(X)) = \frac{2n}{(n+2)^2}\sigma^4$$

- Unbiased but larger variance

$$Var(s_X^2) = \frac{2\sigma^4}{n-1}$$

Lasso/Ridge Regression*

Consider linear regression model $y_i = x_i\beta_0 + u_i$ s.t. $\mathbb{E}[u_i|x_i] = 0$ and $Var(u_i|x_i) = \sigma^2$. In OLS we minimize MSE to estimate β_0, i.e.

$$\beta_0^{OLS} \in \arg\min_{\beta} \mathbb{E}\left[(y_i - x_i\beta)^2\right]$$

Under Gauss-Markov assumptions, we can show that

$$\beta_0^{OLS} = (\sum_{i=1}^{n} x_i^2)^{-1} \sum_{i=1}^{n} x_i y_i$$

is BLUE. However, we can estimate ridge regressor by

$$\beta_0^{Ridge} \in \arg\min_{\beta} \mathbb{E}\left[(y_i - x_i\beta)^2\right] + \lambda\beta^2$$

and get sample estimator

$$\beta_0^{Ridge} = (\sum_{i=1}^{n} x_i^2 + \lambda)^{-1} \sum_{i=1}^{n} x_i y_i$$

Lasso/Ridge Regression* (Contd)

It can be shown that

- Bias $\mathbb{E}\left[\beta_0^{Ridge} - \beta_0 | X\right] = -\frac{\lambda}{\sum_{i=1}^n x_i^2 + \lambda}\beta_0$
- $Var(\beta_0^{Ridge}|X) = \frac{\sum_{i=1}^n x_i^2}{(\sum_{i=1}^n x_i^2 + \lambda)^2}\sigma^2$

  where $Var(\beta_0^{OLS}|X) = \frac{1}{\sum_{i=1}^n x_i^2}\sigma^2$
- When λ depends on n and $\lambda \longrightarrow 0$ when $n \longrightarrow 0$, β_0^{Ridge} is a consistent estimator.